SUPER ADMIN PLATFORM

School Management System
Technical Specification & Implementation Guide — Corrections & Enhancements v1.1

Confidential · Internal Documentation
Audience: Development Team, Platform Engineering, Finance Operations

Related Modules:
Finance · Tenants · Security · Notifications · Analytics

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1. MODULE OVERVIEW & CORRECTION SCOPE

The **Super Admin Platform** acts as the **central control layer of the multi-tenant School Management System SaaS infrastructure**.

It manages:

- Tenant lifecycle
- Platform billing
- Pricing configuration
- Payment processing
- Platform governance
- Security and compliance
- System-wide analytics

This document defines **required corrections and enhancements** to ensure the platform is **ready for production billing activation and financial governance**.

Platform Submodules

Submodule	Description	Priority	Status
Tenant Management	Manage school tenants and lifecycle states	Core	Correction
Subscription & Billing	Manage subscriptions, invoices and payments	Core	Enhancement
Payment Integrations	Payment gateway integration including mobile money	Core	Enhancement
Pricing & Discounts	Pricing configuration, coupons and promotions	Core	Enhancement
Revenue Analytics	SaaS financial performance dashboards	Enhanced	Correction
Platform Settings	Governance, pricing controls and permissions	Core	Correction
Notifications	System alerts and billing communication	Core	Enhancement
Usage Metering	Resource usage monitoring for billing	Enhanced	Enhancement
Security & Compliance	Incident management, audits and backups	Core	Correction
Action Logs	Audit trail for platform actions	Core	Enhancement

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2. TENANT MANAGEMENT

Purpose

The Tenant Management module controls the **lifecycle of all school tenants** within the SaaS platform.

Enhancements ensure that tenant state transitions automatically respond to **subscription billing status**.

Lifecycle States

State	Description
Trial	Tenant using trial plan
Active	Tenant subscription paid
Warning	Payment overdue
Suspended	Access restricted due to unpaid invoices
Cancelled	Tenant subscription cancelled
Archived	Tenant data retained but inactive

Automatic Status Logic

Tenant state should update based on billing events.

Example logic:

```
if invoice.status = paid
    tenant.status = active

if invoice.status = overdue > 7 days
    tenant.status = warning

if invoice.status = overdue > 30 days
    tenant.status = suspended
```

Enhancements

Tenant Financial Profile

Fields added:

Field	Type	Description
plan_id	FK	Current subscription plan
billing_cycle	enum	Monthly / Yearly
trial_end_date	date	Trial expiration
last_payment_date	date	Last successful payment
outstanding_balance	decimal	Pending amount
coupon_applied	FK	Applied coupon code
lifetime_value	decimal	Total revenue generated

API Endpoints

Method	Endpoint	Description
GET	/api/platform/tenants/	List tenants
GET	/api/platform/tenants/{id}/	Tenant details
PATCH	/api/platform/tenants/{id}/status/	Update tenant state
POST	/api/platform/tenants/{id}/suspend/	Suspend tenant
POST	/api/platform/tenants/{id}/activate/	Activate tenant

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3. SUBSCRIPTION & BILLING

Purpose

Provides the **financial engine of the SaaS platform**, responsible for subscription plans, invoices, and payment tracking.

Core Billing Components

Component	Description
Plans	Subscription tiers
Pricing Tables	Plan price configuration
Subscription Manager	Tenant subscription lifecycle
Invoice Engine	Invoice generation and tracking
Payment Processor	Payment gateway integration
Coupon Engine	Discount management

Enhancements

Dynamic Pricing Tables

Pricing must be editable by administrators.

Fields:

Field	Type	Description
plan_name	string	Plan title
monthly_price	decimal	Monthly cost
yearly_price	decimal	Annual cost
student_limit	integer	Maximum students
storage_limit	decimal	Storage quota

Invoice Data Model

Field	Type
id	Integer
tenant	FK
invoice_number	String
amount_due	Decimal
discount_amount	Decimal
net_amount	Decimal
status	enum (Pending / Paid / Overdue)
issued_date	datetime
due_date	datetime

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4. PAYMENT INTEGRATIONS

Purpose

Enable payments through **African mobile money and international payment systems**.

Supported integrations:

- M-Pesa
- Airtel Money
- MTN MoMo
- Flutterwave
- Paystack
- Stripe

Recommended Gateway Architecture

Primary Gateway

```
Flutterwave
```

Fallback Gateway

```
Stripe
```

This configuration allows support for:

- Mobile money
- Card payments
- Bank transfers

Payment API Endpoints

Method	Endpoint
POST	/api/platform/payments/initiate
POST	/api/platform/payments/webhook
GET	/api/platform/payments/history

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5. PRICING, DISCOUNTS & COUPONS

Purpose

Provide flexible pricing and promotional capabilities.

Coupon Types

Type	Example
Percentage	20% discount
Fixed Value	$50 off
Trial Extension	60-day trial
First Payment Discount	50% off first invoice

Coupon Data Model

Field	Type
code	string
discount_type	enum
discount_value	decimal
expiration_date	date
usage_limit	integer
usage_count	Integer
applicable_plans	JSON

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6. REVENUE ANALYTICS

Purpose

Provide financial visibility into SaaS performance.

Key Metrics

Metric
Monthly Recurring Revenue
Annual Recurring Revenue
Average Revenue Per Tenant
Customer Lifetime Value
Churn Rate
Net Revenue
Gross Revenue

Dashboard Views

Revenue breakdown:

- by country
- by plan
- by school size

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7. PLATFORM SETTINGS

Purpose

Acts as the **governance control center** for platform operations.

Corrections Required

Admin Roles

New roles required:

Role
Finance Admin
Billing Manager
Compliance Admin

Role Permissions

Action	Allowed Roles
Edit pricing	Finance Admin
Create coupons	Billing Manager
Process refunds	Finance Admin
Manage tenants	Super Admin

Pricing Version Control

Pricing changes must support rollback.

Fields:

Field	Description
version_number	Pricing revision
changed_by	Admin user
change_reason	Reason for update
effective_date	When pricing takes effect

Notifications Integration

Triggers when:

- pricing changes
- coupon expiry approaching
- plan upgrades/downgrades

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8. USAGE METERING

Purpose

Track platform resource usage for billing.

Metered Resources

Resource
Students
Teachers
Storage
SMS usage
API requests

Example:

```
Plan: Basic
Students included: 500
Extra student cost: $0.10
```

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9. SECURITY & COMPLIANCE

Purpose

Ensure platform governance and regulatory compliance.

Incident Lifecycle

States:

State
Investigating
Resolved
Closed

Severity levels:

Level
Low
Medium
High
Critical

Compliance Reporting

Reports generated:

- monthly
- quarterly
- annual

Reports must reference:

- audit logs
- billing records
- tenant activity

Backup Records

Enhancements:

Feature
Backup verification checks
Timestamped logs
Tenant-level backup control

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10. ACTION LOGS (AUDIT TRAIL)

Every financial and administrative action must be logged.

Logged Events

Event
invoice_created
payment_received
coupon_applied
plan_changed
tenant_suspended

Data Model

Field	Type
id	integer
event_type	string
entity_type	string
entity_id	integer
performed_by	FK(User)
Timestamp	datetime
Metadata	JSON

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11. INTEGRATION POINTS

From	To	Purpose
Billing	Finance	Revenue reporting
Billing	Notifications	Invoice alerts
Tenant	Security	Access control
Usage	Billing	Usage-based charges
Analytics	Dashboard	Platform reporting

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12. PERMISSIONS & ACCESS CONTROL

Module	Super Admin	Finance Admin	Billing Manager
Tenant Management	R/W	R	R
Billing	R/W	R/W	R
Coupons	R/W	R	R/W
Pricing	R/W	R/W	—
Compliance	R/W	R	—